

What Investment Data Implies about the AI Transition

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How much will AI contribute to economic growth?

- Existing answers span orders of magnitude
- Acemoglu (2024): TFP $\uparrow \sim 0.7$ pp over ten years
- Amodei (2024, 2026): “multiple trillions a year by 2028 or 2029” of infrastructure spending; decades of scientific progress in years

Approach: revealed preference from capex data.

1. **Facts.** Five firms; capex \$155B (2022) $\rightarrow$ \$755B (2026 fcst)
2. **Model.** Two-sector economy, AI sector exposed to rare productivity booms
3. **Calibration.** 2024–2027 capex pins down boom size $\hat{\xi} \approx 0.99$
(one boom multiplies AI productivity by $e^{0.99} \approx 2.7$)
4. **Scenarios.** 2028–2029 window of elevated boom probability; aggregate valuations pin down post-window p_*
5. **Implications.** GDP, interest rates, equity premium

Investment facts

Five firms drive AI infrastructure capex

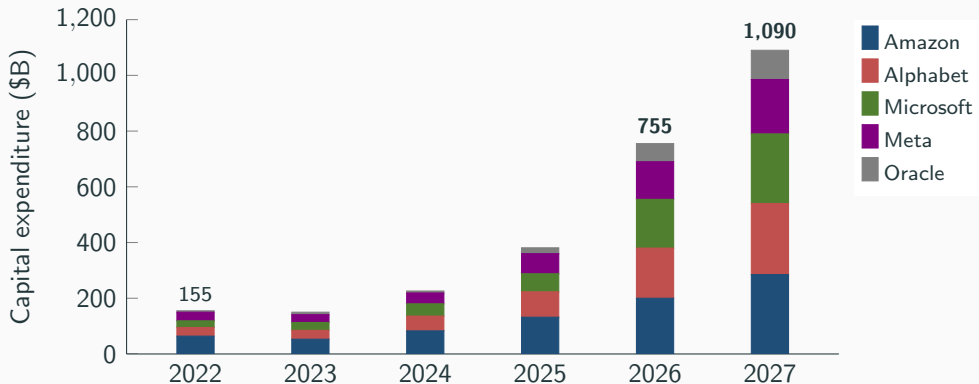
	2022	2023	2024	2025	2026 fcst	2027 est.
Amazon	64	53	83	132	200	285
Alphabet	31	32	53	91	180	255
Microsoft [†]	24	28	44	65	175	250
Meta	31	28	39	72	135	195
Oracle [†]	5	9	7	21	65	105
Total	155	150	226	381	755	1,090

\$ billions. 2022–25: 10-K cash-flow statements ([†]fiscal years ending June, May).

2026: 8-K guidance midpoints; Microsoft from 10-Q actuals and leases not yet commenced.

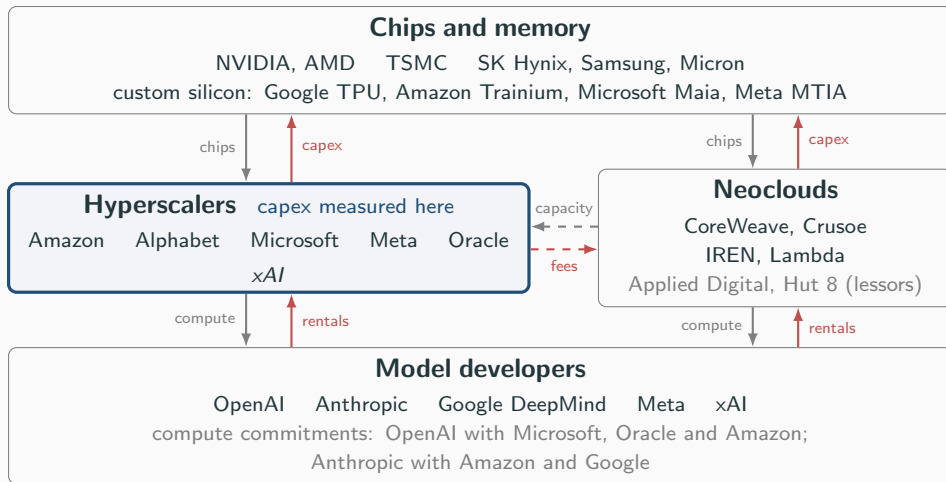
2027: bottom-up from filed commitments. Filing-level sources: paper, Appendix A.

Five firms drive AI infrastructure capex



Flat at about \$150B in 2022–2023; more than tripled between 2024 and 2026.

Where the five firms sit in the AI supply chain



Relative to the macroeconomy

	2022	2024	2025	2026 fcst	2027 est.
AI capex (\$B)	155	226	381	755	1,090
U.S. nominal GDP (\$T)	25.5	29.2	~30.5	~31.4	~32.4
U.S. gross priv. fixed inv. (\$T)	~4.7	~5.2	~5.4	~5.5	~5.7
Share of GDP (%)	0.6	0.8	1.2	2.4	3.4
Share of priv. fixed inv. (%)	3.3	4.3	7.1	13.7	19.2

A two-sector model with rare booms

Two sectors, rare productivity boom

Two sectors $j \in \{a, n\}$ (AI, non-AI), Cobb–Douglas:

$$Y_{jt} = z_{jt}^{1-\alpha} K_{jt}^\alpha, \quad K_{j,t+1} = (1 - \delta_j)K_{jt} + I_{jt}.$$

AI productivity subject to rare boom (probability p_t):

$$z_{a,t+1} = \begin{cases} z_{at} e^\xi & \text{w.p. } p_t \\ z_{at} & \text{w.p. } 1 - p_t \end{cases} \quad \xi > 0.$$

Non-AI unaffected. Timing: $K_{a,t+1}$ chosen at t , before $z_{a,t+1}$ realized.

Firms maximize the present discounted value of earnings at discount rate r_a :

$$V_{at} = \max_{\{I_{as}\}_{s \geq t}} \mathbb{E}_t \sum_{s=t}^{\infty} \frac{1}{(1 + r_a)^{s-t}} [z_{as}^{1-\alpha} K_{as}^\alpha - I_{as}].$$

Frictionless optimal investment

First-order condition w.r.t. $K_{a,t+1}$:

$$\mathbb{E}_t \left[\alpha \left(\frac{z_{a,t+1}}{K_{a,t+1}} \right)^{1-\alpha} \right] = r_a + \delta_a.$$

Define the boom option value:

$$\Omega(p) \equiv (1 - p) + p e^{(1-\alpha)\xi} \geq 1.$$

Optimal capital chosen at time t :

$$K_{a,t+1}^* = z_{at} \left(\frac{\alpha \Omega(p_t)}{r_a + \delta_a} \right)^{1/(1-\alpha)}.$$

Non-AI satisfies same with $\Omega = 1$ (no boom exposure).

Revealed preference: ξ from investment data

Proposition (capital ratio after a boom)

Suppose a boom occurs at t and p shifts $p_{t-1} \rightarrow p_t$. Then

$$\frac{K_{a,t+1}^*}{K_{at}^*} = e^{\xi} \left(\frac{\Omega(p_t)}{\Omega(p_{t-1})} \right)^{1/(1-\alpha)}.$$

Key: the ratio is independent of r_a .

- $r_a + \delta_a$ cancels in the ratio
- $\Rightarrow$ identify ξ from investment data
- $\Rightarrow$ no stand on the required return

Calibration: $\hat{\xi}$ from 2024–2027 capex

Pre-boom AI capital: steady state, $I = \delta_a K$, so

$$K_{a0} = I_{a,\text{pre}}/\delta_a = 155/0.25 = \$620\text{B}.$$

Cumulative 2024–2027 capex: $\bar{I} = 226 + 381 + 755 + 1090 = \$2,452\text{B}.$

Frictionless model: full adjustment in year 1, then replacement of depreciation:

$$\bar{I} = K_{a0}(\delta_a + e^{\xi} + 3\delta_a e^{\xi} - 1)$$

Inverting,

$$e^{\hat{\xi}} = \frac{\bar{I}/K_{a0} + 1 - \delta_a}{1 + 3\delta_a} = \frac{2452/620 + 0.75}{1.75} \approx 2.69, \quad \hat{\xi} \approx 0.99.$$

Assumes zero economic profits and no change in discount rates; $p_{\text{pre}} \approx 0$.

The same calculation firm by firm

Firm	2022 capex (\$B)	2024–27 capex (\$B)	$e^{\hat{\xi}_i}$	$\hat{\xi}_i$
Amazon	64	700	2.0	0.69
Meta	31	441	2.5	0.90
Alphabet	31	579	3.1	1.13
Microsoft	24	534	3.6	1.28
Oracle	5	198	6.1	1.81
Aggregate	155	2,452	2.7	0.99

Each firm as if it were the sole survivor: high on a per-firm basis too.

Alternative interpretation: overoptimism

The revealed-preference $\hat{\xi}$ is what **managers believe**. Two channels drive a wedge to true productivity:

1. **Collective bubble.** Late-1990s fiber-optic precedent — firms invested in capacity not justified by subsequent demand.
2. **Market structure.** Winner-take-all race (Dasgupta–Stiglitz 1980): each firm invests as if it may capture a disproportionate share $\Rightarrow$ aggregate over-investment.

Risk aversion, anchoring, or the quiet life could instead lead to under-investment.
Then the revealed-preference argument is conservative.

Other caveats: pointing both ways

1. **Universe is incomplete.**

↑↑ $\hat{\xi}$

Excludes xAI + neoclouds (CoreWeave, Crusoe, IREN, Lambda).

With xAI: \$795B (2026), \$1,150B (2027); with neoclouds, 2027 $\approx$ **\$1,220B** vs. \$1,090B.

2022 baseline $\approx$ unchanged $\Rightarrow \bar{I}/K_{a0}$ rises $\Rightarrow$ larger $\hat{\xi}$.

2. **Timing: end-load the catch-up.**

↑↑ $\hat{\xi}$

Current formula puts catch-up in 2024 $\Rightarrow$ 3 yrs of depreciation eaten before measurement.

Data are back-loaded (\$226B $\rightarrow$ \$1,090B) $\Rightarrow$ depreciation terms overstated $\Rightarrow$ larger $\hat{\xi}$.

3. **Boom-size vs. boom-probability conflation.**

↓↓ $\hat{\xi}$

Some 2024–27 capex (forward supply agreements, 2027 revenue contracts) reflects option value of *future* booms ($p_{\max}$ window), not the realized 2024 boom. Re-attribution: total commitment unchanged.

Scenario design: a window of elevated p

Structural break in boom probability:

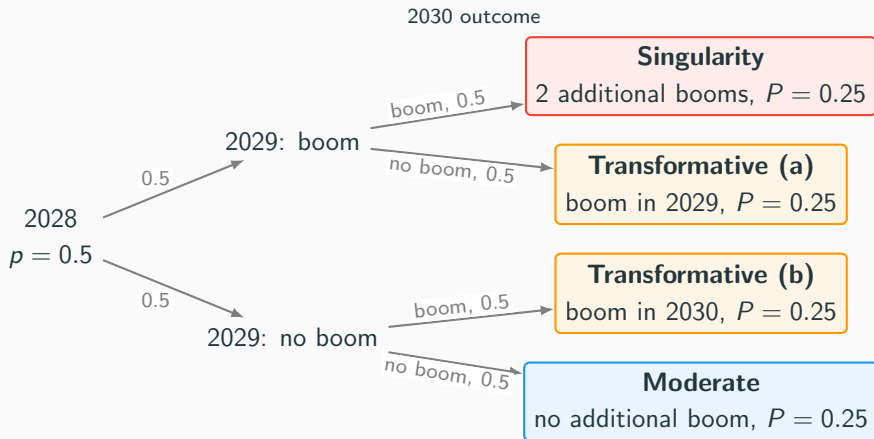
$$p_t = \begin{cases} p_{\text{pre}} \approx 0 & t < 2028 \\ p_{\text{max}} = 0.50 & t \in \{2028, 2029\} \\ p_* & t > 2029 \end{cases}$$

Two i.i.d. draws at $p_{\text{max}} = 0.50$:

Scenario	Add'l booms	Probability
Moderate	0	0.25
Transformative	1	0.50
Singularity	2	0.25

$p_{\text{max}} = 0.5$ gives maximal uncertainty (a Bernoulli variable has maximal variance at 0.5).

Four scenarios from two coin flips



Identifying p_* from aggregate valuations

After the window, p reverts to a long-run value p_* . Discipline from the S&P 500:

- $P/D \approx 94 \Rightarrow D/P \approx 1.06\%$
- 30-yr TIPS yield 2.7% + ERP 5% $\Rightarrow r \approx 8\%$
- Gordon: $D/P = r - g \Rightarrow g \approx 7\%$

Match model-implied AI-sector growth to long-run dividend growth:

$$p_*(e^{\xi} - 1) = g = 0.07 \implies \boxed{p_* \approx 0.04}$$

GDP growth

How a boom translates to output

A boom at $t + 1$: productivity jumps, capital is predetermined $\Rightarrow$ two-stage response.

Impact period ($t + 1$): only the productivity factor moves

$$Y_{a,t+1} = e^{(1-\alpha)\xi} z_{at}^{1-\alpha} (K_{a,t+1}^*)^\alpha.$$

At $\alpha = 1/3$, $\xi = 0.99$: $e^{(1-\alpha)\xi} \approx 1.93$ (+93%).

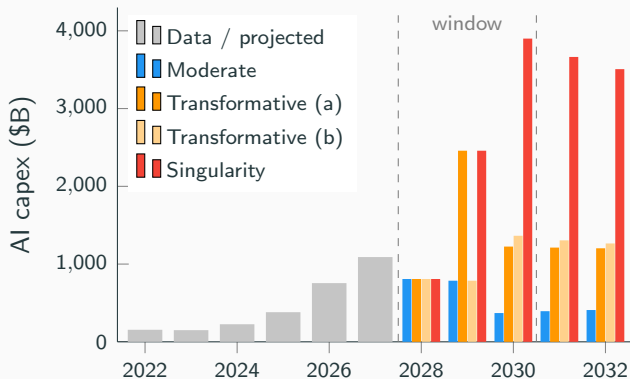
Adjustment period ($t + 2$): capital catches up, $K_{a,t+2}^* = e^\xi K_{a,t+1}^*$

$$Y_{a,t+2} = e^\xi z_{at}^{1-\alpha} (K_{a,t+1}^*)^\alpha.$$

Full factor $e^\xi \approx 2.69$ realized; capital contributes the remaining $e^{\alpha\xi} \approx 1.39$.

α governs the split between immediate and deferred output response.

Investment paths by scenario



Frictionless targets:

- 2028: $p \uparrow p_{\max}$ lifts K^* from \$1,669B to \$2,966B;
 $l_{2028} = \$1,715B$, same in all scenarios
- 2029 boom: $l_{2029} = \$5,758B$;
no boom: \$741B

Partial adjustment from 2028:

$$K_{t+1} = K_t + (K^* - K_t)/\theta_a, \theta_a = 3$$

Frictionless post-window ($\delta_a K^*$):

\$441B / \$1,186B / \$3,192B per year (mod./trans./sing.)

From capital paths to GDP growth

AI output along each scenario path; non-AI output held at its pre-boom level Y_{n0} (no spillovers):

$$Y_{at} = z_{at}^{1-\alpha} K_{at}^{\alpha}, \quad z_{at} = z_0 e^{n_t \xi} \quad (n_t = \text{booms realized by } t, \text{ incl. the initial boom}).$$

Cumulative GDP growth from AI:

$$\frac{Y_{at} + Y_{n0}}{\text{GDP}_0} - 1, \quad \text{GDP}_0 = Y_{a0} + Y_{n0}.$$

AI capital:

- through 2027: $K_{a,t+1} = (1 - \delta_a)K_{at} + I_{at}$, with I_{at} from the capex data
- from 2028: $K_{t+1} = K_t + (1/\theta_a)(K^* - K_t)$, $\theta_a = 3$
- from 2030 (simulations): frictionless, $K_{at} = K^*(z_{at}, p_*)$; booms i.i.d. at $p_* = 0.04$

AI share of output: $s_a = Y_a / (Y_a + Y_n)$.

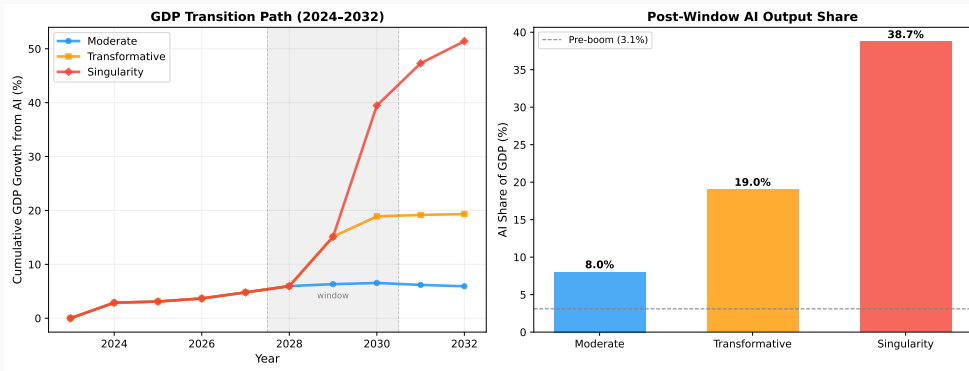
Post-window snapshot

	AI share of output (%)	Cum. GDP growth (%)	Post-window I_a (\$B/yr)
Pre-boom	3.1	—	155
Moderate	8.0	5.4	441
Transformative	19.0	19.7	1,186
Singularity	38.7	58.2	3,192

Steady state at scenario-terminal z_a with $K_a = K_a^*(z_a, p_*)$, no booms after 2030; non-AI output fixed (no spillovers). **Long-run expected AI-sector growth** (aggregate growth from AI: s_a times this):

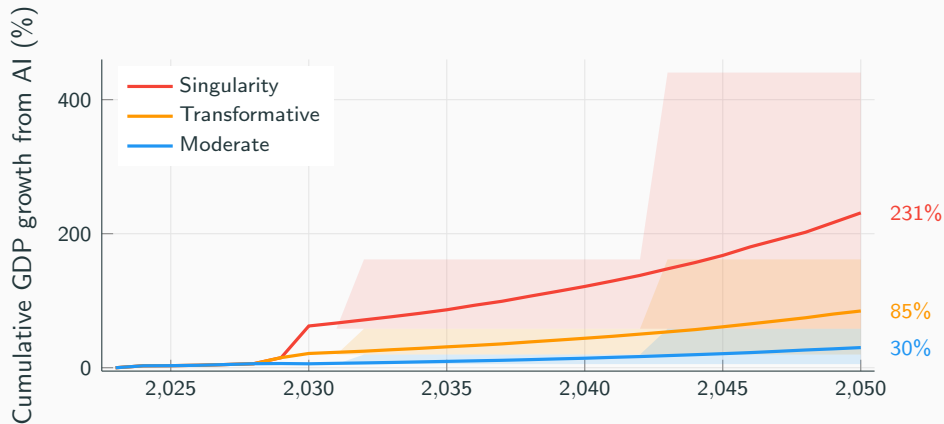
$$g = p_*(e^\xi - 1) = 0.04 \times 1.69 \approx 6.8\%/yr, \quad \text{in rare discrete jumps.}$$

GDP transition path (2024–2032)



Moderate $\approx$ 5% cumulative; Transformative $\approx$ 20%; Singularity $\approx$ 58%.

Simulated paths to 2050 with ongoing booms at $p_* = 0.04$



Lines: means over 10,000 paths. Bands: 10th–90th percentiles. Frictionless capital from 2030 (hence the step).

Three observations about the long run

1. **The moderate scenario is already large.** One boom and nothing further: ~ 5 pp by 2030, ~ 30 pp in expectation by 2050.
2. **With no spillovers, the AI share is first order.** Aggregate growth from AI is the sector's growth weighted by its share; 8% vs. 39% post-window is what compounds into 30% vs. 231% by 2050.
3. **Once the share is large, Bernoulli uncertainty dominates everything else.** Singularity 10th–90th range by 2050: 58% to 440%.

Asset-pricing implications

From GDP to consumption

Representative agent (Epstein–Zin, EIS = 1, risk aversion γ) prices an exogenous consumption stream:

$$\Delta c_{t+1} = \mu + \sigma \varepsilon_{t+1} + x_{d,t+1} + x_{b,t+1}, \quad x_b = \xi_b \text{ w.p. } p_b = p_* = 0.04$$

Disasters: $\xi_d = -0.15$ w.p. $p_d = 0.036$ (Barro 2006; Wachter 2013). $\mu = 0.025$, $\sigma = 0.02$, $\rho = 0.02$.

The sector boom reaches consumption through the AI share: $\xi_b = s_a (1 - \alpha) \xi$

	Moderate	Transformative	Singularity	Absorbing limit
s_a	0.080	0.190	0.387	1
ξ_b	0.053	0.126	0.256	0.660

- $(1 - \alpha)\xi$ is the impact-period output response; s_a weights it by the sector's size
- Post-window s_a is a lower bound (the sector keeps growing); $s_a = 1$ is the absorbing limit; both understate the long-run effect on consumption

Equilibrium rates of return

	No AI	Moderate	Transformative	Singularity	Absorbing limit
p_b	0	0.04	0.04	0.04	0.04
ξ_b	—	0.053	0.126	0.256	0.660
<i>Risk-free rate r_f (%)</i>					
$\gamma = 1$	3.90	4.11	4.37	4.81	5.85
$\gamma = 3$	3.63	3.81	4.00	4.18	4.16
$\gamma = 5$	3.29	3.46	3.58	3.63	3.44
<i>Equity risk premium (%)</i>					
$\gamma = 1$	0.12	0.13	0.18	0.37	1.84
$\gamma = 3$	0.39	0.42	0.55	1.00	3.53
$\gamma = 5$	0.72	0.77	0.97	1.55	4.25

At $\gamma = 3$: r_f up 18–55 bp; equity premium up 3 bp (moderate) to 314 bp (absorbing limit).

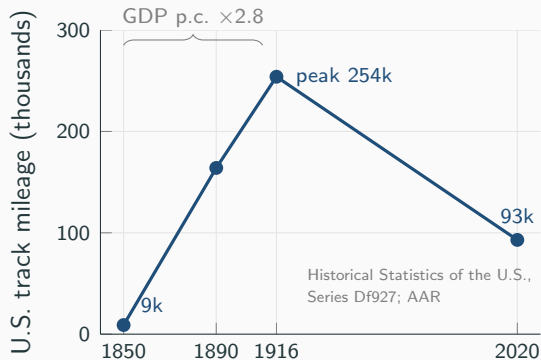
Historical context and conclusion

Growth episodes on one scale

Episode	Period	Years	Multiplier	
			GDP	AI sector
British Industrial Revolution	1760–1840	80	1.7	—
U.S. railroad era	1850–1910	60	2.8	—
Second Industrial Revolution (U.S.)	1870–1920	50	2.4	—
Japan postwar; Korea; Taiwan; Singapore; China	1945–2020	25–30	8–13	—
U.S. IT boom	1995–2005	10	1.5	—
AI moderate (1 boom)	2024–2030	6	1.22	2.7
AI transformative (2 booms)	2024–2030	6	1.39	7.2
AI singularity (3 booms)	2024–2030	6	1.83	19.5
AI singularity, post-window, 30 years	2030–2060	30	7.1	7.1

Historical rows: GDP per capita, PPP (Maddison 2007 and the Maddison Project Database; Barro 2006; Penn World Table 10.0; Broadberry et al. 2015 for Britain). AI rows compound the share-weighted AI contribution with 2.5%/yr trend growth (trend alone: $1.16\times$ over 2024–30).

The railroad analogue: overshooting came decades after the gain



1. Track 9,000 $\rightarrow$ 254,000 miles; GDP per capita $\times 2.8$ over the railroad era
2. 63% of peak mileage abandoned, but decades later, with the rise of competing transportation
3. The productivity gain was real; the decline was a later shock to the sector's z
4. The question is whether the boom justifies the investment when made: at $\delta_a = 0.25$, today's hardware is largely replaced within four years

What if managers are wrong?

The revealed-preference $\hat{\xi}$ is what managers *believe*. Wedges to the true gain:

True gain smaller

$\hat{\xi}$ too high

- **Mirage / bubble.** Late-1990s fiber precedent; customers will not pay for more AI usage $\Rightarrow$ NPV-negative capex, defaults, disinvestment
- **Misattributed p .** Capex that anticipates a rise in p is read as boom size: $\hat{\xi}$ too high, eventual capital unchanged

True gain larger

$\hat{\xi}$ too low

- **Under-investment.** Managerial risk aversion, anchoring, the quiet life
- **Option to buy distressed rivals' capital** below replacement cost reduces the incentive to build now
- Plus the measurement items: five-firm scope, straight-line depreciation, zero profits

Winner-take-all races (Dasgupta–Stiglitz 1980): $\hat{\xi}$ unaffected (per firm, $\hat{\xi}_i = 0.69\text{--}1.81$); only the eventual level of capital is lower.

What investment data adds

- Task-based estimates: TFP ~ 0.7 pp over ten years (Acemoglu 2024)
- Capex commitments: AI-sector productivity $\times 2.7$, and 5 pp of GDP by 2030 with no further booms and no spillovers
- Wages and financing adjust without frictions: the wage clears each sector's labor market, and global investors supply capital at r_a

Conclusion

Boom size from 2024–27 capex (\$2.45T)	$e^{\hat{\xi}} \approx 2.7$	open economy, r_a exogenous
Cumulative GDP gain by 2030	5–58 pp	partial equilibrium,
AI share of output after the window	8–39%	no spillovers
Long-run expected AI-sector growth	$\approx 7\%/yr$, in rare jumps	aggregate rate, absorbing limit
Risk-free rate ($\gamma = 3$)	+18–55 bp	equilibrium,
Equity premium ($\gamma = 3$)	+3 bp to +3 pp	investment given

If the boom fails to materialize, this is the largest capital misallocation in U.S. history.
Those certain of that outcome may be making the 2007 forecasters' mistake in reverse.

Rare events are hard to imagine until they occur.